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### Location coordinate-based user authentication with device loss security tolerance

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#### Abstract

According to one embodiment, a method, computer system, and computer program product for user authentication. The embodiment may include receiving, at an authentication server, respective multiple location coordinates from first and second registered devices. Storing, on the authentication server, respective moving windows comprising registered, device specific, last n location coordinates from the first and the second registered devices. Receiving a request to access the authentication server via the first registered device. The request comprises log-in credentials of a user and a first hash value calculated at the first registered device. Computing, at the authentication server, a second hash value using a set of the last n location coordinates from each of the respective moving windows. Comparing the first hash value and the second hash value. In response to the first and the second hash values being equal, and the log-in credentials being verified, granting access to the authentication server.

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## **Background/Summary**

### **BACKGROUND**

(1) The present invention relates generally to the field of computing, and more particularly to multi-factor authentication of a user.

(2) User authentication verifies the identity of a user requesting access to a computing system by authorizing a human-to-machine transfer of credentials during interactions on a network to confirm the user's authenticity. Authentication helps to ensure only authorized users gain access (i.e., log-in) to a network or computing system by preventing unauthorized users from gaining such access. Categories of factors (i.e., pieces of evidence) used in user authentication may include knowledge factors, possession factors, and inherence factors. Knowledge factors may include things the user must know in order to log-in to the network or computing system (e.g., username, user id, password, personal identification numbers). Possession factors may include things the user must have in their possession in order to complete the log-in (e.g., one-time password (OTP), key fob, smartphone app, employee identification card). Inherence factors may include characteristics inherent to the user which confirm their identity (e.g., biometrics, facial recognition, voice authentication). Multi-factor authentication (MFA) is an electronic authentication method in which the user is granted access only after successfully presenting factors from two or more categories to an authentication mechanism. For example, the user may be required to provide a user id and password in combination with an OTP in order to log-in to a computing system.

### **SUMMARY**

(3) According to one embodiment, a method, computer system, and computer program product for user authentication. The embodiment may include receiving, at an authentication server, respective multiple location coordinates from a first registered device and a second registered device. The embodiment may include storing, on the authentication server, respective moving windows comprising registered, device specific, last n location coordinates of the multiple location coordinates from the first registered device and the second registered device. The embodiment may include receiving a request to access the authentication server via the first registered device. The request comprises log-in credentials of a user and a first hash value calculated at the first registered

device. The embodiment may include computing, at the authentication server, a second hash value based on a set of the last n location coordinates from each of the respective moving windows. The embodiment may include verifying the log-in credentials of the user. The embodiment may include comparing the first hash value and the second hash value. In response to the first hash value and the second hash value being equal, and the log-in credentials being verified, the embodiment may include granting access to the authentication server via the first registered device.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) These and other objects, features and advantages of the present invention will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings. The various features of the drawings are not to scale as the illustrations are for clarity in facilitating one skilled in the art in understanding the invention in conjunction with the detailed description. In the drawings:

(2) FIG. 1 illustrates an exemplary computer environment according to at least one embodiment.

(3) FIG. 2 illustrates an operational flowchart for authenticating a user via a multi-device location sampling multi-factor authentication process according to at least one embodiment.

### DETAILED DESCRIPTION

(4) Detailed embodiments of the claimed structures and methods are disclosed herein; however, it can be understood that the disclosed embodiments are merely illustrative of the claimed structures and methods that may be embodied in various forms. This invention may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.

(5) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces unless the context clearly dictates otherwise.

(6) The present invention relates generally to the field of computing, and more particularly to multi-factor authentication of a user. The following described exemplary embodiments provide a system, method, and program product to, among other things, authenticate a user via a received hash value of sampled (i.e., identified) location coordinates of two registered computing devices of the user, in addition to received log-in credentials (e.g., user id and password) of the user.

Therefore, the present embodiment has the capacity to improve the technical field of multi-factor authentication by authenticating a user based, in part, on a dynamically computed hash value of two sets of identified location coordinates of a user which is sent to an authenticating server along with log-in credentials of the user, thus eliminating multiple round-trip communications between the user and the authenticating server. Further, the present embodiment has the capacity to improve the technical field of multi-factor authentication by restricting access to the authenticating server where a set of identified location coordinates is missing.

(7) As previously described, user authentication verifies the identity of a user requesting access to a computing system by authorizing a human-to-machine transfer of credentials during interactions on a network to confirm the user's authenticity. Authentication helps to ensure only authorized users gain access (i.e., log-in) to a network or computing system by preventing unauthorized users from gaining such access. Categories of factors (i.e., pieces of evidence) used in user authentication may include knowledge factors, possession factors, and inherence factors. Knowledge factors may include things the user must know in order to log-in to the network or computing system (e.g., username, user id, password, personal identification numbers). Possession factors may include things the user must have in their possession in order to complete the log-in (e.g., one-time

password (OTP), key fob, smartphone app, employee identification card). Inherence factors may include characteristics inherent to the user which confirm their identity (e.g., biometrics, facial recognition, voice authentication). MFA is an electronic authentication method in which the user is granted access only after successfully presenting factors from two or more categories to an authentication mechanism. For example, the user may be required to provide a user id and password in combination with an OTP in order to log-in to a computing system.

(8) An OTP is a password that is valid for only one log-in session or transaction on a computer system or other digital device. OTPs may provide more enhanced protection than static passwords which remain the same for multiple log-in sessions as an OTP works through randomness algorithms that generate a new and random password each time they are used. OTP generation may utilize several techniques including time-synchronization and mathematical algorithm schemes. The OTP mechanism to enhance authentication is widely used, however, it is time consuming as multiple round-trip communications between a user and an authenticating server are required before access is granted. For example, a first round-trip communication between the user and the authenticating server may include the user sending, via a registered device, a user id and password to the authenticating server and, in response, receiving, at the registered device, an OTP from the authenticating server. Subsequently, a second round-trip communication between the user and the authenticating server may include the user sending, via the registered device, the OTP back to the authenticating server and, in response, receiving verification/access from the authenticating server. These multiple round-trip communications between the registered device of the user and the authenticating server and may delay the user as they wait to receive an OTP and may consequently reduce productivity of the user (e.g., when the user is prevented from accessing resources needed in real-time). Further, as this MFA mechanism relies on the registered device being in possession of the user, access security of the authenticating server may be compromised in situations where the registered device is not in possession of the user (e.g., the registered device has been lost, misplaced, or stolen), but rather, is in possession of another individual. It may therefore be imperative to have a system in place to provide MFA utilizing information relating to identified location coordinates from two registered devices of a user which is sent for authentication, along with a user id and a password of the user, thereby securing access to an authenticating server when one of the registered devices is not in possession of the user. Thus, embodiments of the present invention may be advantageous to, among other things, compute a hash value of respective identified location coordinates of a first computing device and a second computing device of a user which are both co-located with the user, authenticate the user utilizing the computed hash value in combination with a user id and a password of the user, eliminate the need for a second round-trip communication for MFA of the user, and prevent access to computing resources of an authenticating server where one of the first or the second computing device is not within the user's possession. The present invention does not require that all advantages need to be incorporated into every embodiment of the invention.

(9) According to at least one embodiment, a user may register two (i.e., a first and a second) Global Positioning System (GPS) enabled electronic devices (e.g., a laptop and a smartphone) with an authenticating server. The first and the second registered devices may periodically identify their respective location coordinates and share them with the authenticating server. The first and the second registered devices may also communicate with each other via short range communication (e.g., Bluetooth). A respective moving window of location coordinates (e.g., the last  $n$  identified location coordinates) may be maintained on each of the two registered devices. Further, the respective moving windows of location coordinates of the two registered devices may also be stored on the authenticating server. Upon the user requesting access to the authenticating server via one of the registered devices, login credentials (e.g., user id and password) of the user and a registered device computed hash value of sets of location coordinates from the respective moving windows of location coordinates of the two registered devices may be sent to the authenticating

server for authentication. The user may be granted access to the authenticating server where the authentication has determined valid login credentials and a match between the registered device computed hash value and an authenticating server computed hash value of the sets of location coordinates from the respective moving windows of location coordinates of the two registered devices. According to at least one embodiment, a frequency of location coordinate identification, a number of location coordinates stored within respective moving windows, and an applied hashing algorithm may be defined during a system initialization phase and implemented by both of the registered devices as well as the authenticating server. According to at least one other embodiment, access to the authenticating server may be based, in part, on a comparison of the respective identified location coordinates of both registered devices maintained on the registered device requesting access and the respective identified location coordinates of both registered devices maintained on the authenticating server. According to at least one further embodiment, an OTP may be additionally utilized for authentication where there has been no change among the respective last  $n$  identified location coordinates of the two registered devices.

(10) Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

(11) A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random-access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

(12) The following described exemplary embodiments provide a system, method, and program product to authenticate a user based, in part, on information relating to respective sets of identified location coordinates of two electronic devices co-located with the user and, accordingly, grant the user access to a computing system, and/or an application of the computing system, in response to a successful authentication.

(13) Referring to FIG. 1, an exemplary computing environment **100** is depicted, according to at least one embodiment. Computing environment **100** contains an example of an environment for the

execution of at least some of the computer code involved in performing the inventive methods, such as authentication program **107A**, **107B**, **107C**. In addition to authentication program **107A**, **107B**, **107C**, computing environment **100** includes, for example, computer **101**, wide area network (WAN) **102**, end user device (EUD) **103**, remote server **104**, public cloud **105**, and private cloud **106**. In this embodiment, computer **101** includes processor set **110** (including processing circuitry **120** and cache **121**), communication fabric **111**, volatile memory **112**, persistent storage **113** (including operating system **122**, software program **108**, and authentication program **107A**), peripheral device set **114** (including user interface (UI) device set **123**, storage **124**, and Internet of Things (IoT) sensor set **125**), and network module **115**. Remote server **104** includes remote database **130**. Public cloud **105** includes gateway **140**, cloud orchestration module **141**, host physical machine set **142**, virtual machine set **143**, and container set **144**.

(14) Computer **101** may take the form of a desktop computer, laptop computer, tablet computer, smartphone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **130**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **100**, detailed discussion is focused on a single computer, specifically computer **101**, to keep the presentation as simple as possible. Computer **101** may be located in a cloud, even though it is not shown in a cloud in FIG. **1**. On the other hand, computer **101** is not required to be in a cloud except to any extent as may be affirmatively indicated.

(15) Processor set **110** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **120** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **120** may implement multiple processor threads and/or multiple processor cores. Cache **121** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **110**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set **110** may be designed for working with qubits and performing quantum computing.

(16) Computer readable program instructions are typically loaded onto computer **101** to cause a series of operational steps to be performed by processor set **110** of computer **101** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **121** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **110** to control and direct performance of the inventive methods. In computing environment **100**, at least some of the instructions for performing the inventive methods may be stored in authentication program **107A**, **107B**, **107C** in persistent storage **113**, end user device **103**, and remote server **104**, respectively.

(17) Communication fabric **111** is the signal conduction paths that allow the various components of computer **101** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

(18) Volatile memory **112** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM.



Typically, the volatile memory is characterized by random access, but this is not required unless affirmatively indicated. In computer **101**, the volatile memory **112** is located in a single package and is internal to computer **101**, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer **101**.

(19) Persistent storage **113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid-state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface type operating systems that employ a kernel. The code included in authentication program **107A**, **107B**, **107C** typically includes at least some of the computer code involved in performing the inventive methods.

(20) Peripheral device set **114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as smart glasses and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (for example, where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer, another sensor may be a motion detector, and yet another sensor may be a GPS receiver which provides computer **101** with satellite navigation and geographic location identification capability.

(21) Network module **115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from an external computer or external storage device through a network adapter card or network interface included in network module **115**.

(22) WAN **102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN may be replaced and/or

supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

(23) End user device (EUD) **103** is any computer system that is used and controlled by an end user (for example, a client of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** may be capable of hosting and running an authentication program **107B** and communicating with computer **101** via WAN **102**, in accordance with embodiments of the invention. Authentication program **107B** may be the same as or substantially similar to authentication program **107A**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

(24) Remote server **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**, and may take any of the forms discussed above in connection with computer **101**. Remote server **104** may be capable of hosting and running an authentication program **107C** and communicating with computer **101** via WAN **102**, in accordance with embodiments of the invention. Authentication program **107C** may be the same as or substantially similar to authentication program **107A**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

(25) Public cloud **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

(26) Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can

utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

(27) Private cloud **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

(28) The authentication program **107A**, **107B**, **107C** may be a program capable of granting a user access to a computing system based on, in part, authentication of data relating to periodically identified respective location coordinates of two registered computing devices of the user. In at least one embodiment, the authentication program **107A**, **107B**, **107C** may require a user to opt-in to system usage upon opening or installation of the authentication program **107A**, **107B**, **107C**. The authentication method is explained in further detail below with respect to FIG. 2. Notwithstanding depiction in computer **101**, end user device **103**, and remote server **104**, authentication program **107A**, **107B**, **107C** may be stored in and/or executed by, individually or in any combination, computer **101**, end user device **103**, remote server **104**, public cloud **105**, and private cloud **106** so that functionality may be separated among the devices.

(29) Referring now to FIG. 2, an operational flowchart for authenticating a user via a multi-device location sampling/identifying multi-factor authentication process **200** is depicted according to at least one embodiment. At **202**, authentication program **107A**, **107B**, **107C** periodically receives respective GPS location coordinates of two registered electronic devices (e.g., a laptop and a smartphone) which are co-located with a user. According to least one embodiment, during an initialization/set-up phase of authentication program **107A**, **107B**, **107C** the user may register a first electronic device (e.g., computer **101**) and a second electronic device (e.g., EUD **103**), either of which the user may utilize to access another computing device (e.g., remote server **104**), and/or an application of the other computing device, pending a successful authentication by authentication program **107A**, **107B**, **107C**. Each electronic device (e.g., computer **101** and EUD **103**) registered by the user may execute a respective local instance of authentication program **107A**, **107B**, **107C** and may utilize respective GPS receivers to periodically identify their respective geographic location coordinates (i.e., numeric values for latitude and longitude). A location coordinate identification frequency for registered devices may be initially specified by the user during the set-up phase of authentication program **107A**, **107B**, **107C** and may be later updated by the user or by authentication program **107A**, **107B**, **107C**. For example, the user may initially specify a location coordinate identification frequency of once every half hour, hour, day, etc. for all registered devices. During implementation, the registered devices may periodically identify, according to the location coordinate identification frequency, their respective geographic location coordinates and transmit them to an instance of authentication program **107A**, **107B**, **107C** executing on another computing device such as an authentication server (e.g., remote server **104**). Furthermore, authentication program **107A**, **107B**, **107C** may instruct the registered devices to locally store a respective moving window of its last n identified location coordinates (e.g., the last four identified location coordinates). The size of the moving window (i.e., the value of n) may be initially specified by the user during the set-up phase of authentication program **107A**, **107B**, **107C** and may be later updated by the user or by authentication program **107A**, **107B**, **107C**.

(30) Next, at **204**, the authentication program **107A**, **107B**, **107C** stores, on the authentication server (e.g., remote server **104**), respective moving windows of the location coordinates received for the first and the second registered devices (e.g., computer **101** and EUD **103**). According to at least one embodiment, the stored moving windows of respective location coordinates received from the registered devices may each be equal in size (i.e., the value of *n*) to the respective moving windows of identified location coordinates stored on both of the registered devices. For example, the first and the second registered devices may each store the last four of their respectively identified location coordinates within their locally stored moving window, and the authentication server may also locally store two moving windows, one for each of the first and the second registered devices. Further, each locally stored moving window on the authentication server may contain the last four identified location coordinates received from a respective registered device. In doing so, the moving window stored on the authentication server for the first registered device may be identical to the moving window stored on the first registered device (i.e., they each contain the same set of identified location coordinates for the first registered device), and similarly, the moving window stored on the authentication server for the second registered device may be identical to the moving window stored on the second registered device (i.e., they each contain the same set of identified location coordinates for the second registered device).

(31) At **206**, the authentication program **107A**, **107B**, **107C** receives a request from the user to access the authentication server, and/or an application on the authentication server, via the first registered device. According to at least one embodiment, the received request from the user may include log-in credentials of the user, such as a user id and a password. Furthermore, the received user request to access the authentication server may trigger the instance of the authentication program **107A**, **107B**, **107C** executing on the first registered device to retrieve a set of identified location coordinates from the moving window of the last *n* identified location coordinates stored on the second registered device, as described below at **208**.

(32) At **208**, the authentication program **107A**, **107B**, **107C** determines whether a set of identified location coordinates of the second registered device has been received at the first registered device from the second registered device. According to at least one embodiment, the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device and the instance authentication program **107A**, **107B**, **107C** executing on the second registered device may only communicate with each other via a short/limited range wireless communication protocol (e.g., Bluetooth, infrared, or near-field communication). Utilization of such a limited range communication protocol enhances authentication security as it necessitates a limited proximity between the first registered device and the second registered device. According to at least one embodiment, the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device may communicate, via Bluetooth, with the second registered device and obtain a set of identified location coordinates of the second registered device.

(33) Additionally, at **208**, the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device may calculate a hash value of a set of identified location coordinates from its locally stored moving window (i.e., the last *n* identified location coordinates of the first registered device) and from the obtained set of identified location coordinates of the second registered device. For example, where the locally stored moving window includes the last four identified location coordinates of the first registered device and the obtained set includes the last four identified location coordinates of the second registered device, the hash value may be computed using a set of the last one, two, three, or four identified location coordinates from each of the locally stored moving window on the first registered device and the obtained set of identified location coordinates of the second registered device. The hash value may be requested on-demand by the user, via the first registered device, and computed by the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device. Alternatively, the hash value may be automatically computed by the instance of authentication program **107A**, **107B**, **107C** executing on

the first registered device in response to the user's request to access the authentication server. The calculated hash value and the log-in credentials of the user may be transmitted to the authentication server as part of the user's request. Moreover, the hashing algorithm (H) (e.g., a version of secure hash algorithm (SHA) or message-digest algorithm (MD5)) utilized by authentication program **107A, 107B, 107C** may be initially specified by the user during the set-up phase of authentication program **107A, 107B, 107C** and may be later updated by the user or by authentication program **107A, 107B, 107C**. Also, the number of identified location coordinates to be hashed (L) may be initially specified by the user during the set-up phase of authentication program **107A, 107B, 107C** and may be later updated by the user or by authentication program **107A, 107B, 107C**.

(34) According to at least one other embodiment, the instance of authentication program **107A, 107B, 107C** executing on the first registered device may calculate a first hash value of a numbered set of identified location coordinates from its locally stored moving window (i.e., the last n identified location coordinates of the first registered device), and may also calculate a second hash value of a same numbered set of identified location coordinates from the obtained set of identified location coordinates of the second registered device. For example, where the locally stored moving window includes the last four identified location coordinates of the first registered device and the obtained set includes the last four identified location coordinates of the second registered device, the first hash value may be computed using a set of the last one, two, three, or four identified location coordinates from the locally stored moving window on the first registered device, and second hash value may be computed using a same numbered set of the last identified location coordinates from the obtained set of identified location coordinates of the second registered device. The first and the second hash values may be automatically computed by the instance of authentication program **107A, 107B, 107C** executing on the first registered device in response to the user's request to access the authentication server. Further, the first and the second calculated hash values, along with the log-in credentials of the user, may be transmitted to the authentication server as part of the user's request.

(35) According to at least one further embodiment, the instance of authentication program **107A, 107B, 107C** executing on the first registered device may communicate, via Bluetooth, with the second registered device and obtain a hash value of a set of identified location coordinates from the moving window of the last n identified location coordinates stored on the second registered device. For example, where the moving window includes the last four identified location coordinates of the second registered device, the hash value may be computed using a set of the last one, two, three, or four identified location coordinates from the moving window stored on the second registered device. The hash value may be requested on-demand by the user, via the first registered device, and computed by the instance of authentication program **107A, 107B, 107C** executing on the second registered device. Alternatively, the hash value may be automatically computed by the instance of authentication program **107A, 107B, 107C** executing on the second registered device in response to the instance of authentication program **107A, 107B, 107C** executing on the first registered device requesting a set of identified location coordinates of the second registered device. Correspondingly, the instance of authentication program **107A, 107B, 107C** executing on the first registered device may calculate a second hash value of a same numbered set of identified location coordinates from its locally stored moving window. Further, the obtained hash value and the second hash value, along with the log-in credentials of the user, may be transmitted to the authentication server as part of the user's request. Rather than determining whether a set of identified location coordinates of the second registered device has been received at the first registered device, the authentication program **107A, 107B, 107C** may determine whether a hash value of a set of identified location coordinates from the moving window stored on the second registered device has been received at the first registered device.

(36) With continued reference to step **208**, in response to determining that a set of identified location coordinates of the second registered device has been received at the first registered device

(step **208**, “Y” branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **210**. In response to determining that a set of identified location coordinates of the second registered device has not been received at the first registered device (step **208**, “N” branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **214**.

(37) Next, at **210**, the authentication program **107A**, **107B**, **107C** determines whether authentication (i.e., validation) of the received user request for access to the authentication server is successful or not. In making this determination, the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server (e.g., remote server **104**) may extract the user id and password from the received request and compare them against log-in credentials (e.g., user ids and passwords) stored in a database (e.g., remote database **130**) of authorized users of the authentication server. Furthermore, the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server may also compute a hash value of a set of identified location coordinates from the respective moving windows of the location coordinates received from the first and the second registered devices and stored on the authentication server. Both the hashing algorithm (H) and the number of identified location coordinates to be hashed (L) utilized by the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server may be the same H and L utilized by the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device. For example, the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device may utilize the hashing algorithm MD5 to compute a hash value based on the last three identified location coordinates within its locally stored moving window and the last three identified location coordinates from the obtained set of identified location coordinates of the second registered device. Similarly, the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server may also utilize the hashing algorithm MD5 to compute a hash value based on the last three identified location coordinates within its locally stored moving window for the first registered device and the last three identified location coordinates within its locally stored moving window for the second registered device.

(38) With continued reference to step **210**, authentication of the received user request for access to the authentication server may be determined successful where there is both a match between the log-in credentials (i.e., user id and password) received at **206** and log-in credentials stored in the database of authorized users of the authentication server, and a match (i.e., equal values) between the hash value computed at **208** by the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device (i.e., the hash value of a set of identified location coordinates from its locally stored moving window and from the obtained set of identified location coordinates of the second registered device) and the hash value computed by the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server. It should be noted that any difference between the respective location coordinates utilized by the instances of authentication program **107A**, **107B**, **107C** executing on the first registered device and on the authentication server to compute their respective hash values may result in the respective hash values not being equal. Authentication of the received user request for access to the authentication server may not be determined successful where there is not a match between the hash value computed by the instance of authentication program **107A**, **107B**, **107C** executing on the first registered device and the hash value computed by the instance of authentication program **107A**, **107B**, **107C** executing on the authentication server. In response to determining that authentication of the received user request for access to the authentication server is successful (step **210**, “Y” branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **212**. In response to determining that authentication of the received user request for access to the authentication server is not successful (step **210**, “N” branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **214**.

(39) According to at least one other embodiment, determination of whether authentication of the

received user request for access to the authentication server is successful may be based, in part, on a comparison of identified location coordinates of the respective moving windows of location coordinates received from the first and the second registered devices and stored on the authentication server against identified location coordinates of the first and the second registered devices transmitted from the first registered device to the authentication server as part of the user's request. In such an embodiment, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may compare the last n identified location coordinates of each of the registered devices, which were received from the first registered device as part of the user's request, to the last n identified location coordinates of the respective moving windows for the first and the second registered devices stored on the authentication server. For example, authentication of the received user request for access to the authentication server may be determined successful where there is a match between the log-in credentials (i.e., user id and password) received at **206** and log-in credentials stored in the database of authorized users of the authentication server, and where the identified location coordinates maintained on the authentication server for both the first and the second registered devices match the identified location coordinates of the first registered device and the second registered device which were transmitted from the first registered device to the authentication server as part of the user's request. Authentication of the received user request for access to the authentication server may be determined unsuccessful where the identified location coordinates maintained on the authentication server for both the first and the second registered devices differ from the identified location coordinates of the first registered device and the second registered device which were transmitted from the first registered device to the authentication server as part of the user's request. In response to determining that authentication of the received user request for access to the authentication server is successful (step **210**, "Y" branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **212**. In response to determining that authentication of the received user request for access to the authentication server is not successful (step **210**, "N" branch), the multi-device location sampling/identifying multi-factor authentication process **200** may proceed to step **214**.

(40) According to yet another embodiment, determination of whether authentication of the received user request for access to the authentication server is successful may be based, in part, on a comparison of a first hash value and a second hash value received from the first registered device as part of the user's request and a first hash value and a second hash value computed by the instance of authentication program **107A, 107B, 107C** executing on the authentication server. As described above, an embodiment may exist where the instance of authentication program **107A, 107B, 107C** executing on the first registered device calculates a first hash value of a numbered set of identified location coordinates from its locally stored moving window, calculates a second hash value of a same numbered set of identified location coordinates from the obtained set of identified location coordinates of the second registered device, and transmits the first and the second hash values to the authentication server as part of the user's request. In such an embodiment, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may calculate a first hash value of a same numbered set of identified location coordinates from its locally stored moving window for the first registered device and may also calculate a second hash value of a same numbered set of identified location coordinates from its locally stored moving window for the second registered device. Authentication of the received user request for access to the authentication server may be determined successful where there is a match between the log-in credentials received at **206** and log-in credentials stored in the database of authorized users of the authentication server, a match (i.e., equal values) between the first hash value received from the first registered device and the first hash value calculated by the instance of authentication program **107A, 107B, 107C** executing on the authentication server, and a match between the second hash value received from the first registered device and the second hash value calculated by the instance

of authentication program **107A, 107B, 107C** executing on the authentication server.

(41) According to a further embodiment, where there is no change among two or more identified location coordinates within a respective moving window of the location coordinates received for the first registered device or the second registered device and stored on the authentication server, authentication program **107A, 107B, 107C** may additionally utilize, or revert to, an OTP during the determination of whether authentication of the received user request for access to the authentication server is successful. For example, if the last two or more of the last  $n$  identified location coordinates received for the first registered device and stored within a moving window of the authentication server are the same (i.e., there has been no change), the instance of authentication program **107A, 107B, 107C** executing on the authentication server may additionally utilize an OTP procedure as part of the determination of whether authentication of the received user request for access to the authentication server is successful.

(42) At **212**, in response to determining that authentication of the received user request for access to the authentication server is successful, the instance of authentication program **107A, 107B, 107C** executing on the authentication server grants the user access to the authentication server, and/or an application on the authentication server, via the first registered device. Accordingly, through the use of the multi-device location sampling/identifying multi-factor authentication process **200**, the user has gained access to the authentication server based on a close proximity the first registered device and the second registered device and after one round-trip of communication between the first registered device and the authentication server despite the implementation of multi-factor authentication.

(43) According to at least one other embodiment, in response to determining that authentication of the received user request for access to the authentication server is successful, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may also determine if an update to one or more of the location coordinate identification frequency ( $F$ ), the size ( $N$ ) of the respective moving windows maintained on the first and second registered devices and on the authentication server, the hashing algorithm ( $H$ ) utilized by the first and second registered devices and the authentication server, and the number of identified location coordinates to be hashed ( $L$ ) using  $H$  is required. As mentioned above, the values of  $F$ ,  $N$ ,  $H$ , and  $L$  may be initially configured by the user during a set-up phase and implemented uniformly across the instances of authentication program **107A, 107B, 107C** executing on the first and second registered devices and on the authentication server. Post set-up, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may unilaterally determine that an update is required to one or more of  $F$ ,  $N$ ,  $H$ , and  $L$ , and transmit an update message to the instances of authentication program **107A, 107B, 107C** executing on the first and second registered devices while currently, or before, granting the user access to the authentication server, and/or an application on the authentication server. The update message may contain information specifying new values for one or more of  $F$ ,  $N$ ,  $H$ , and  $L$  to be implemented uniformly across all executing instances of authentication program **107A, 107B, 107C**. The instance of authentication program **107A, 107B, 107C** executing on the authentication server may update one or more of its respective  $F$ ,  $N$ ,  $H$ , and  $L$  upon transmitting the update message, and the instances of authentication program **107A, 107B, 107C** executing on the first and second registered devices may update one or more of their respective  $F$ ,  $N$ ,  $H$ , and  $L$  upon receiving the update message.

(44) For example, a required change to the location coordinate identification frequency ( $F$ ) may be determined where location coordinates of the first and the second registered devices are being identified frequently (e.g., every 30 minutes), yet the authentication server access request (i.e., log-in) frequency of the user is infrequent (e.g., once every 3 days). In such a scenario, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may determine that a reduction to the location coordinate identification frequency is required. As another example, a required change to the hashing algorithm ( $H$ ) utilized by the first and the second registered



devices and the authentication server may be determined where H has been used beyond a threshold amount of time (e.g., a number of hours or days). As yet another example, a required change to the number of identified location coordinates to be hashed (L) using H may be determined as a consequence of user movement behavior. For instance, in a scenario where the user is not moving frequently and a current L value results in unchanging location coordinates being hashed, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may determine that an increase to L is required.

(45) At **214**, in response to determining that a set of identified location coordinates of the second registered device has not been received at the first registered device, or in response to determining that authentication of the received user request for access to the authentication server is not successful, the instance of authentication program **107A, 107B, 107C** executing on the authentication server denies the user access to the authentication server, and/or an application on the authentication server, via the first registered device. A determination that a set of identified location coordinates of the second registered device has not been received at the first registered device may indicate that the first registered device is outside a communication range with the second registered device, possibly due to a misplacement or a theft of the first registered device. As such, denial of user access to the authentication server via the first registered device is warranted in order to maintain secure access to the authentication server. Furthermore, the instance of authentication program **107A, 107B, 107C** executing on the authentication server may also send a message to the first registered device indicting the reason(s) for the denial of access (e.g., a mismatch between received log-in credentials and log-in credentials stored in a database of authorized users, a mismatch between calculated hash values, a mismatch between identified location coordinates for the first and the second registered devices, or lack of obtained location coordinates of the second registered device). Although not depicted, the multi-device location sampling/identifying multi-factor authentication process **200** may terminate at **214**.

(46) It may be appreciated that FIG. 2 provides only an illustration of one implementation and does not imply any limitations with regard to how different embodiments may be implemented. Many modifications to the depicted environments may be made based on design and implementation requirements.

(47) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

## Claims

1. A computer-implemented method, the method comprising: receiving, at an authentication server, multiple location coordinates from a first registered device; receiving, at the authentication server, multiple location coordinates from a second registered device, wherein the first registered device and the second registered device are separate devices which are both co-located with a user; storing, on the authentication server, respective moving windows comprising registered, device specific, last n location coordinates of the multiple location coordinates from the first registered device and the multiple location coordinates from the second registered device, wherein n is two or more; receiving a request to access the authentication server via the first registered device, wherein the request comprises log-in credentials of the user and a first hash value calculated at the first registered device; computing, at the authentication server, a second hash value based on a set of the last n location coordinates from each of the respective moving windows; verifying the log-in

credentials of the user; comparing the first hash value and the second hash value; and in response to the first hash value and the second hash value being equal, and the log-in credentials being verified, granting access to the authentication server via the first registered device.

2. The method of claim 1, further comprising: storing, on the first registered device and the second registered device, respective moving windows comprising registered device specific last n location coordinates of the respective multiple location coordinates, wherein location coordinates of the respective multiple location coordinates are identified and sent to the authentication server according to a location coordinate identification frequency.

3. The method of claim 2, wherein receiving the request to access the authentication server via the first registered device further comprises: receiving, at the first registered device from the second registered device, a set of identified location coordinates of the second registered device, wherein the first registered device and the second registered device communicate with each other via a short-range wireless communication protocol; and calculating, at the first registered device, the first hash value based on a set of last n location coordinates from each of the respective moving window stored on the first registered device and the received set of identified location coordinates of the second registered device.

4. The method of claim 3, wherein the first hash value and the second hash value are computed using a same hashing algorithm, and wherein the set of the last n location coordinates used to calculate the first hash value and the set of the last n location coordinates used to calculate the second hash value have a same set size, and wherein a set size comprises a number of location coordinates.

5. The method of claim 1, wherein verifying the log-in credentials of the user further comprises: extracting a user id and a password from the log-in credentials; comparing the user id and the password against user ids and passwords stored in a database of authorized users of the authentication server; and determining a match for the user id and the password.

6. The method of claim 1, further comprising: denying access to the authentication server via the first device in response to a determination that a set of identified location coordinates of the second registered device has not been received at the first registered device from the second registered device.

7. The method of claim 1, wherein the first registered device and the second registered device are registered with the authentication server, and wherein the respective multiple location coordinates comprise Global Positioning System (GPS) coordinates.

8. A computer system, the computer system comprising: one or more processors, one or more computer-readable memories, one or more computer-readable tangible storage medium, and program instructions stored on at least one of the one or more tangible storage medium for execution by at least one of the one or more processors via at least one of the one or more memories, wherein the computer system is capable of performing a method comprising: receiving, at an authentication server, multiple location coordinates from a first registered device; receiving, at the authentication server, multiple location coordinates from a second registered device, wherein the first registered device and the second registered device are both separate devices co-located with a user; storing, on the authentication server, respective moving windows comprising registered, device specific, last n location coordinates of the multiple location coordinates from the first registered device and the multiple location coordinates from the second registered device, wherein n is two or more; receiving a request to access the authentication server via the first registered device, wherein the request comprises log-in credentials of the user and a first hash value calculated at the first registered device; computing, at the authentication server, a second hash value based on a set of the last n location coordinates from each of the respective moving windows; verifying the log-in credentials of the user; comparing the first hash value and the second hash value; and in response to the first hash value and the second hash value being equal, and the log-in credentials being verified, granting access to the authentication server via the first registered

device.

9. The computer system of claim 8, further comprising: storing, on the first registered device and the second registered device, respective moving windows comprising registered device specific last  $n$  location coordinates of the respective multiple location coordinates, wherein location coordinates of the respective multiple location coordinates are identified and sent to the authentication server according to a location coordinate identification frequency.

10. The computer system of claim 9, wherein receiving the request to access the authentication server via the first registered device further comprises: receiving, at the first registered device from the second registered device, a set of identified location coordinates of the second registered device, wherein the first registered device and the second registered device communicate with each other via a short-range wireless communication protocol; and calculating, at the first registered device, the first hash value based on a set of last  $n$  location coordinates from each of the respective moving window stored on the first registered device and the received set of identified location coordinates of the second registered device.

11. The computer system of claim 10, wherein the first hash value and the second hash value are computed using a same hashing algorithm, and wherein the set of the last  $n$  location coordinates used to calculate the first hash value and the set of the last  $n$  location coordinates used to calculate the second hash value have a same set size, and wherein a set size comprises a number of location coordinates.

12. The computer system of claim 8, wherein verifying the log-in credentials of the user further comprises: extracting a user id and a password from the log-in credentials; comparing the user id and the password against user ids and passwords stored in a database of authorized users of the authentication server; and determining a match for the user id and the password.

13. The computer system of claim 8, further comprising: denying access to the authentication server via the first device in response to a determination that a set of identified location coordinates of the second registered device has not been received at the first registered device from the second registered device.

14. The computer system of claim 8, wherein the first registered device and the second registered device are registered with the authentication server, and wherein the respective multiple location coordinates comprise Global Positioning System (GPS) coordinates.

15. A computer program product, the computer program product comprising: one or more computer-readable tangible storage medium and program instructions stored on at least one of the one or more tangible storage medium, the program instructions executable by a processor capable of performing a method, the method comprising: receiving, at an authentication server, multiple location coordinates from a first registered device; receiving, at the authentication server, multiple location coordinates from a second registered device, wherein the first registered device and the second registered device are both separate devices co-located with a user; storing, on the authentication server, respective moving windows comprising registered, device specific, last  $n$  location coordinates of the multiple location coordinates from the first registered device and the multiple location coordinates from the second registered device, wherein  $n$  is two or more; receiving a request to access the authentication server via the first registered device, wherein the request comprises log-in credentials of the user and a first hash value calculated at the first registered device; computing, at the authentication server, a second hash value based on a set of the last  $n$  location coordinates from each of the respective moving windows; verifying the log-in credentials of the user; comparing the first hash value and the second hash value; and in response to the first hash value and the second hash value being equal, and the log-in credentials being verified, granting access to the authentication server via the first registered device.

16. The computer program product of claim 15, further comprising: storing, on the first registered device and the second registered device, respective moving windows comprising registered device specific last  $n$  location coordinates of the respective multiple location coordinates, wherein location

coordinates of the respective multiple location coordinates are identified and sent to the authentication server according to a location coordinate identification frequency.

17. The computer program product of claim 16, wherein receiving the request to access the authentication server via the first registered device further comprises: receiving, at the first registered device from the second registered device, a set of identified location coordinates of the second registered device, wherein the first registered device and the second registered device communicate with each other via a short-range wireless communication protocol; and calculating, at the first registered device, the first hash value based on a set of last n location coordinates from each of the respective moving window stored on the first registered device and the received set of identified location coordinates of the second registered device.

18. The computer program product of claim 17, wherein the first hash value and the second hash value are computed using a same hashing algorithm, and wherein the set of the last n location coordinates used to calculate the first hash value and the set of the last n location coordinates used to calculate the second hash value have a same set size, and wherein a set size comprises a number of location coordinates.

19. The computer program product of claim 15, wherein verifying the log-in credentials of the user further comprises: extracting a user id and a password from the log-in credentials; comparing the user id and the password against user ids and passwords stored in a database of authorized users of the authentication server; and determining a match for the user id and the password.

20. The computer program product of claim 15, further comprising: denying access to the authentication server via the first device in response to a determination that a set of identified location coordinates of the second registered device has not been received at the first registered device from the second registered device.

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